



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

Instructor: Prashant Kumar Pandey



Azure
Databricks

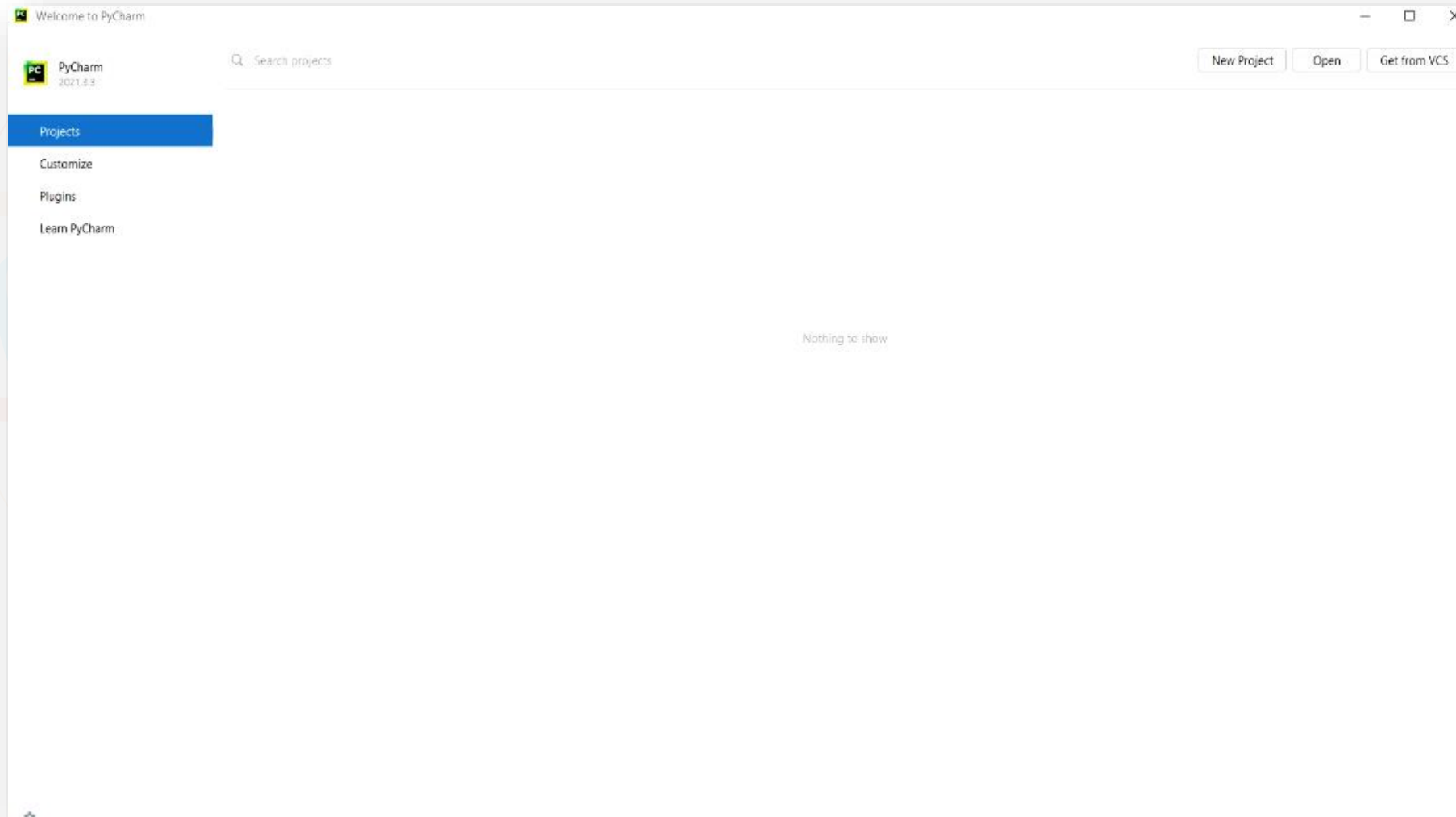


Absolute Beginner to Specialization in Apache Spark and Azure Databricks

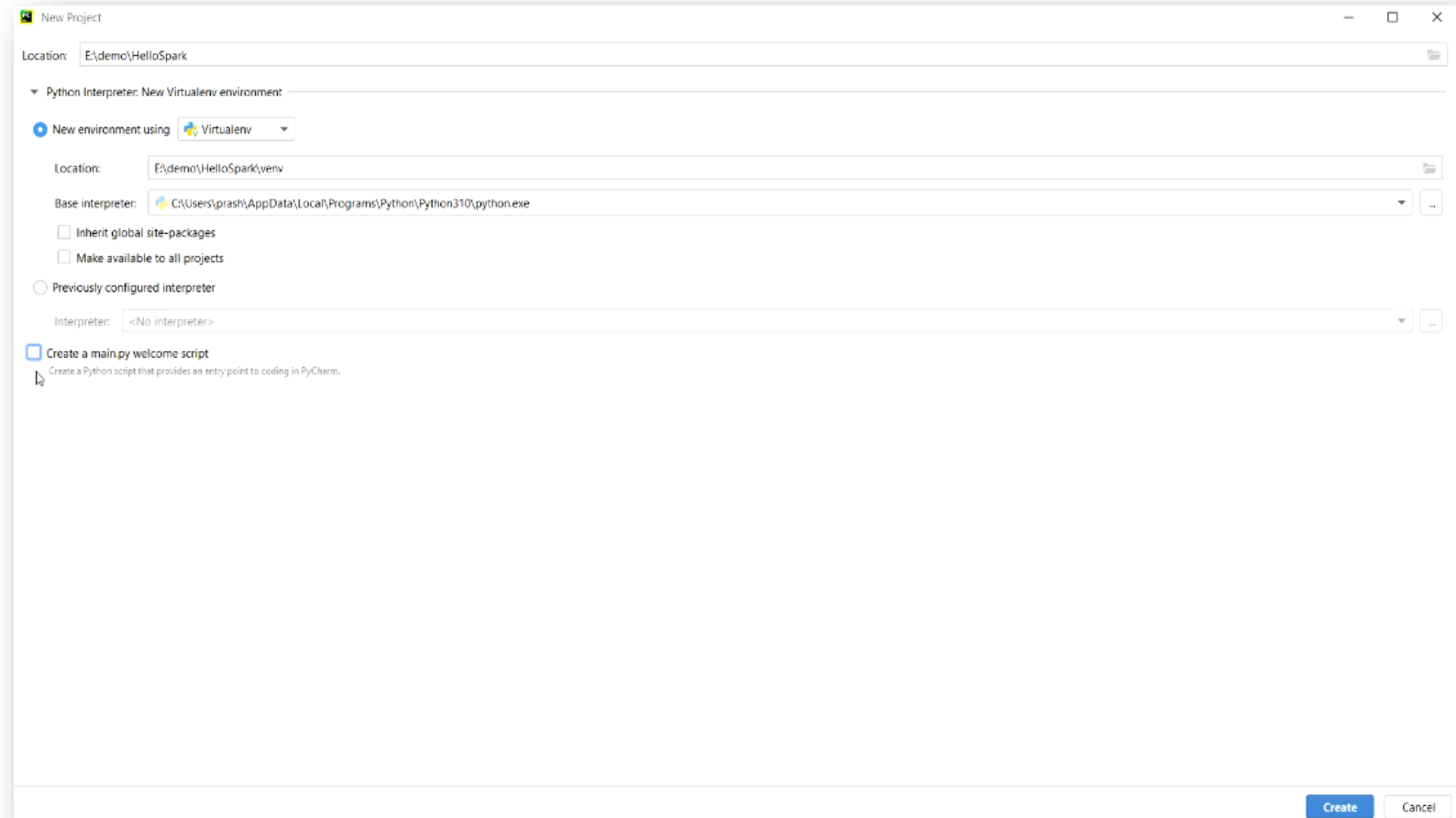


First Spark Application using IDE

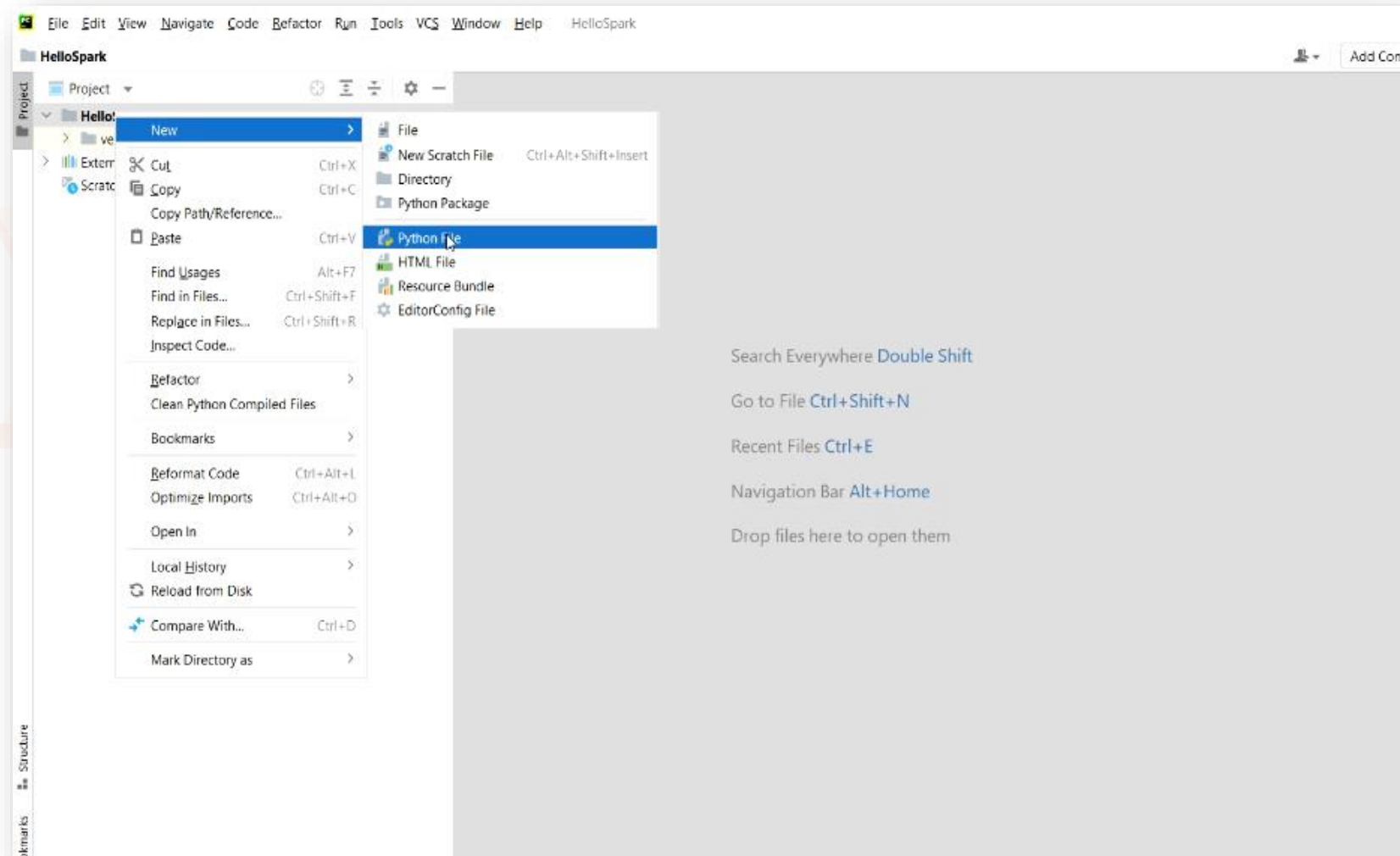
Start your PyCharm IDE and create a new project.



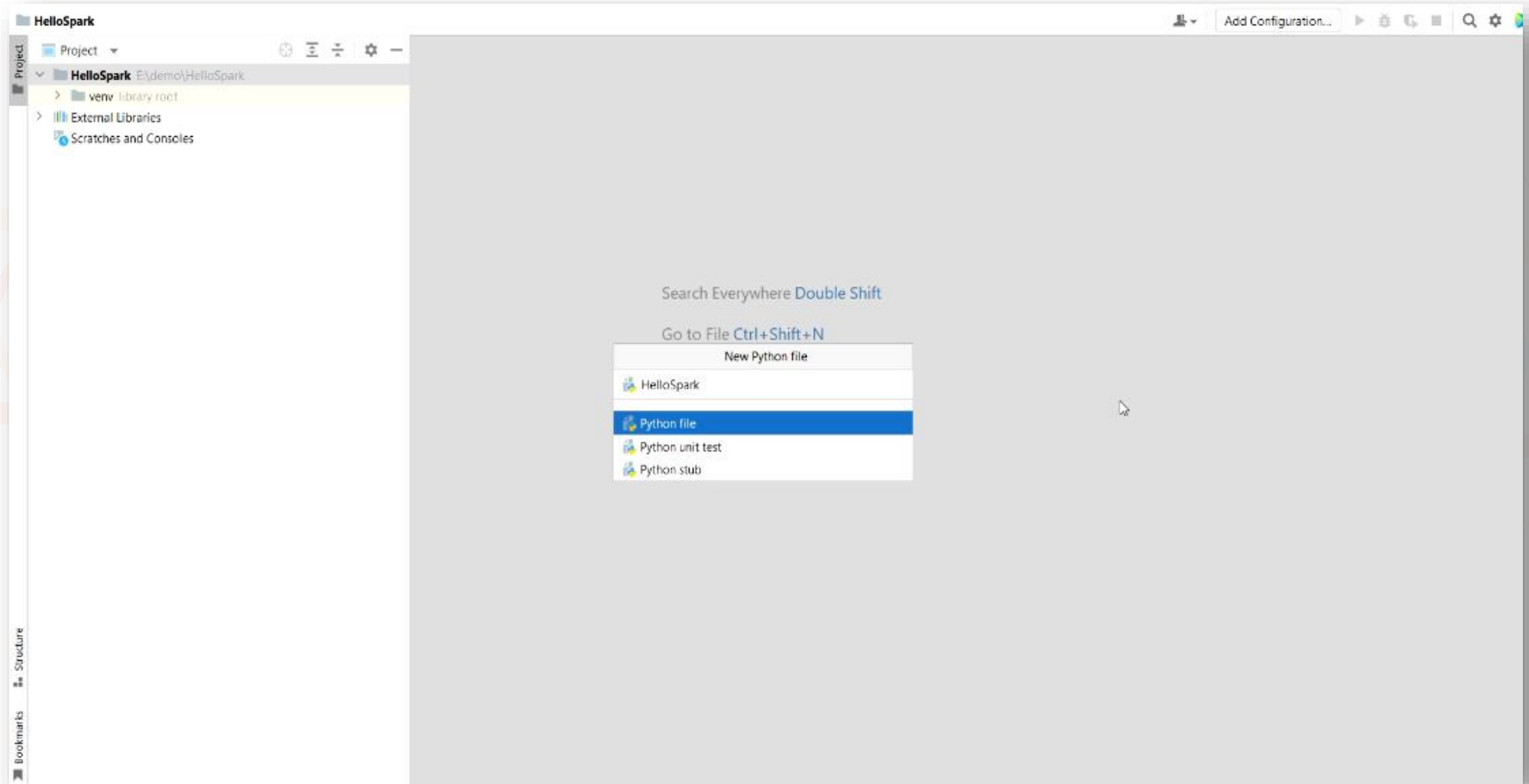
1. Set your new project location by hitting the browse button and choosing your working directory. Also type a new project directory location.
2. Ensure you have the latest version of Python selected in the Python Interpreter dropdown.
3. Uncheck all other options. We do not need anything else.
4. Use VirtualEnv for setting up our project.
5. Hit the create button.



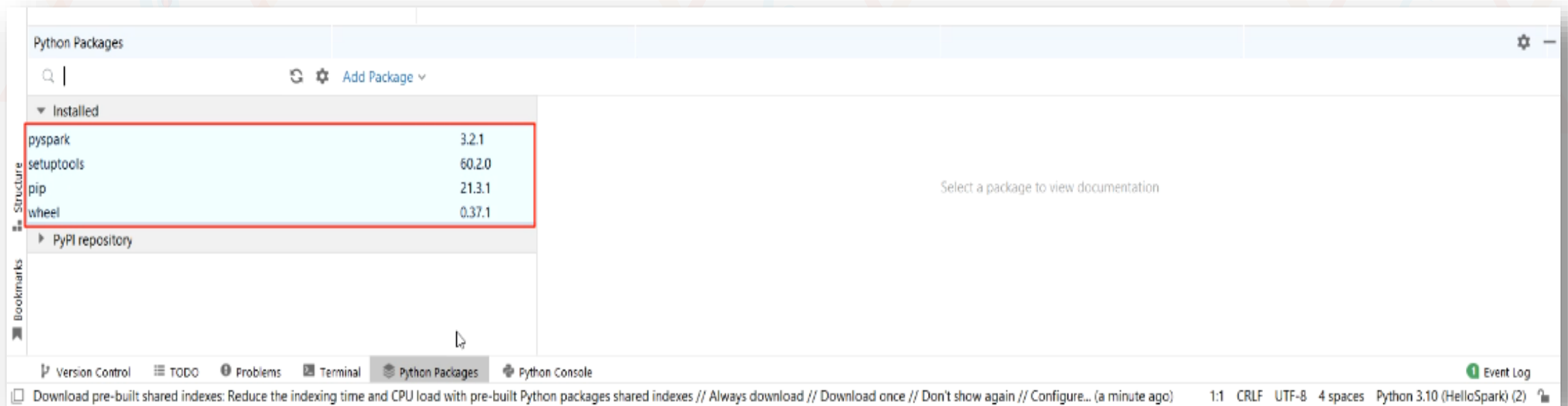
Once your blank project setup is done. You can create a new Python file in your project directory as shown in the image below.



Give a name to your Python file.



Make sure you have the PySpark package installed in your project's VirtualEnv. You check that by clicking the python packages window at the bottom of your IDE. And you should see your installed packages there as shown in the image below. I already have the pyspark package installed.



However, if you do not see PySpark in the list of your installed packages, search for it. Select the pyspark package under the PyPI repository. Come to the right side, three dots, and you should see an option to install.

I see a delete option here because I have installed it already.

But you may see an install button to install the selected package.

Install it if you do not have it already installed.

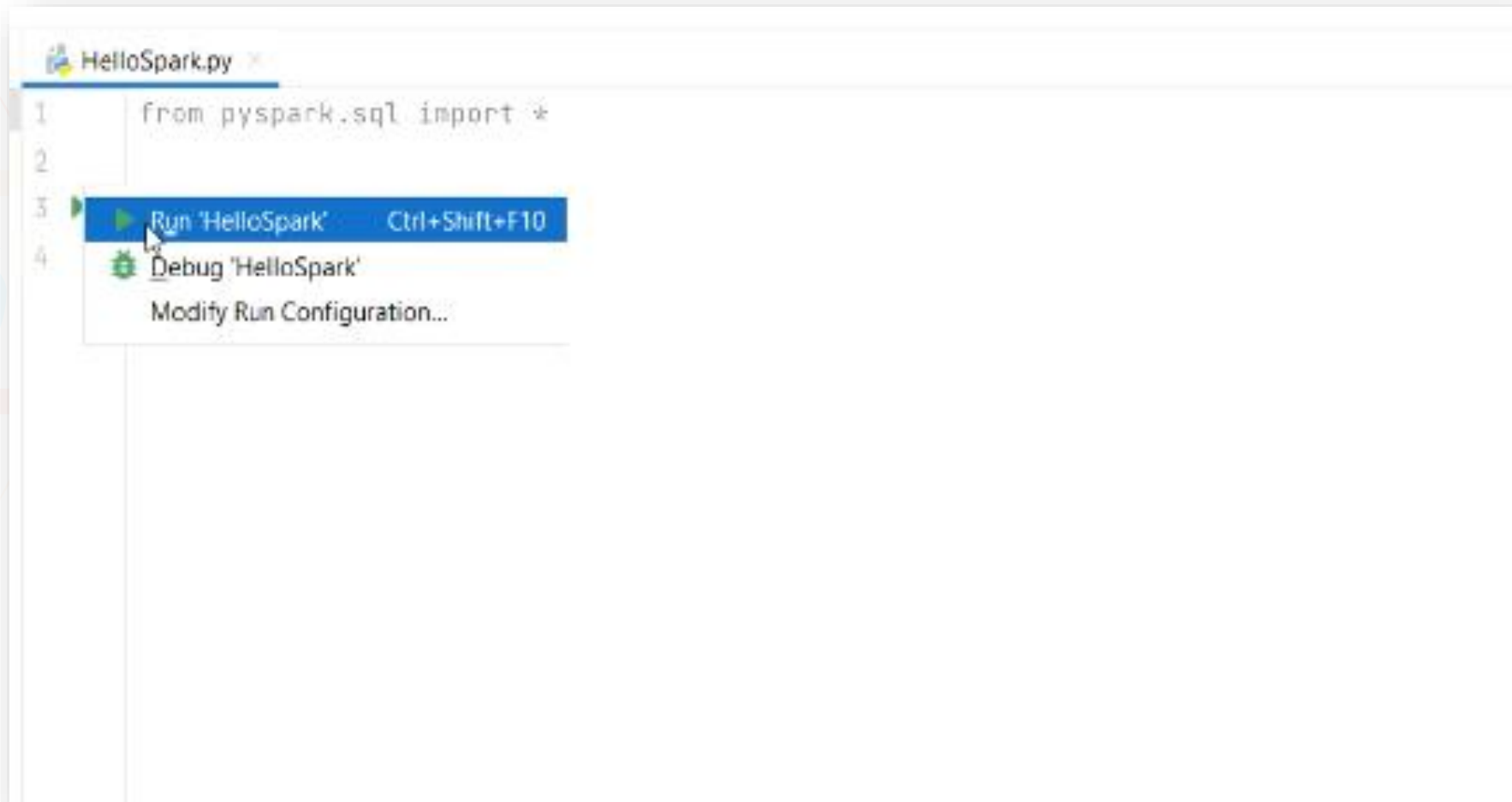


We haven't started learning Spark programming yet. So I am not expecting you to understand the code. However, I want to create a super simple Spark application and run it from the IDE. In the screenshot shown below, we have a simple Hello Spark Python code, it is not a Spark program.

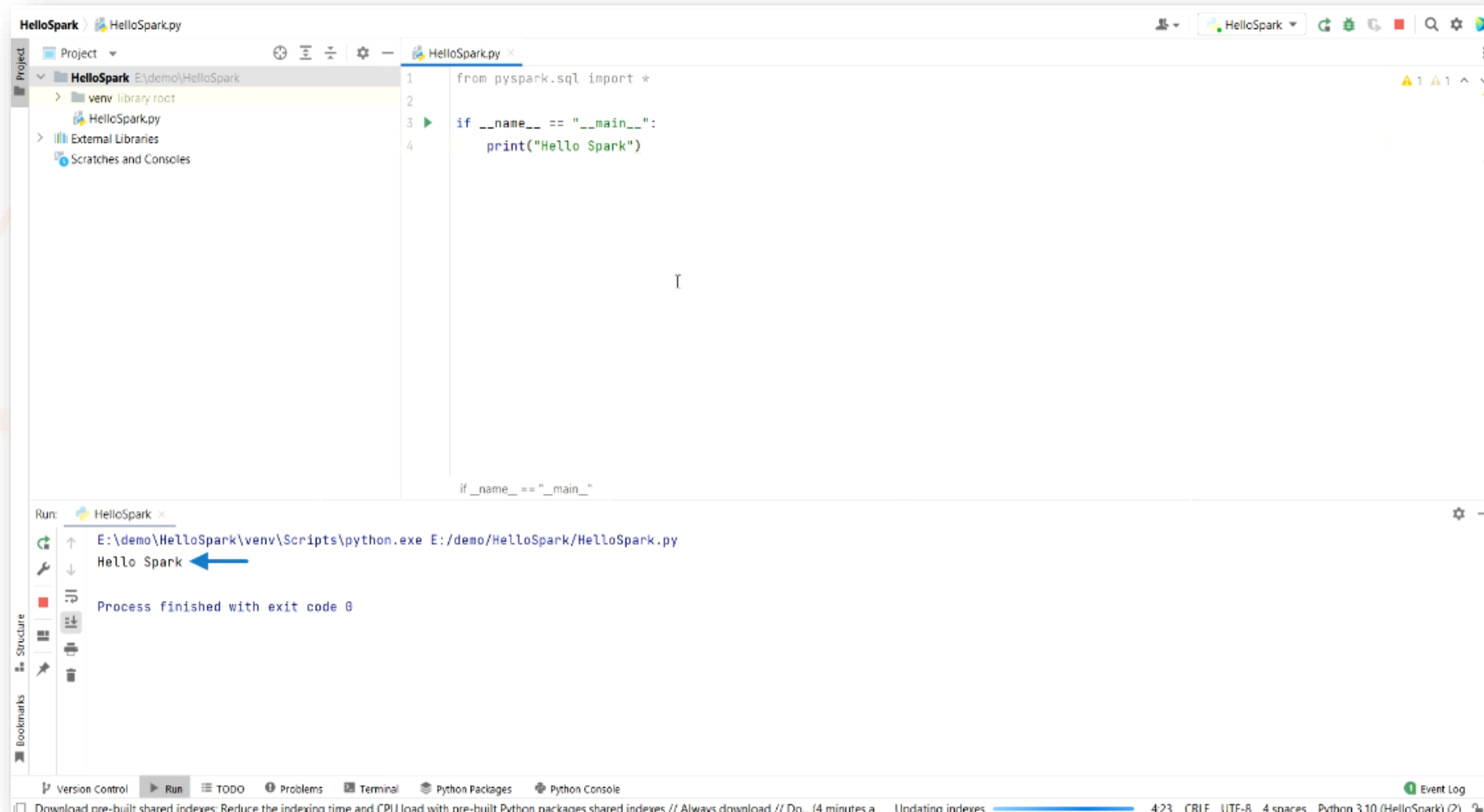


```
1 from pyspark.sql import *
2
3 if __name__ == "__main__":
4     print("Hello Spark")
```

Run your program as shown below.



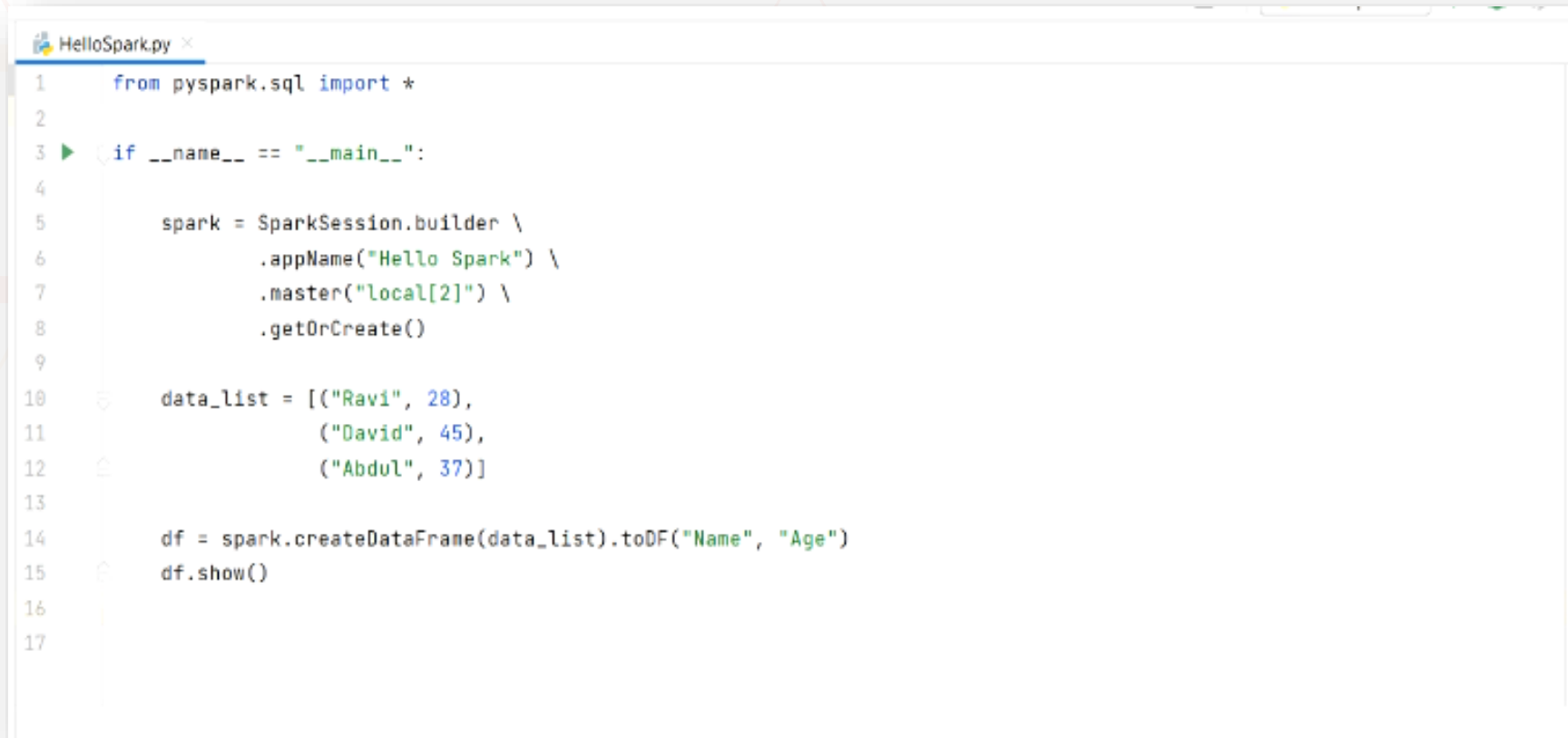
It worked! I can see the output in the console below.



Now, let us look at a Spark code.

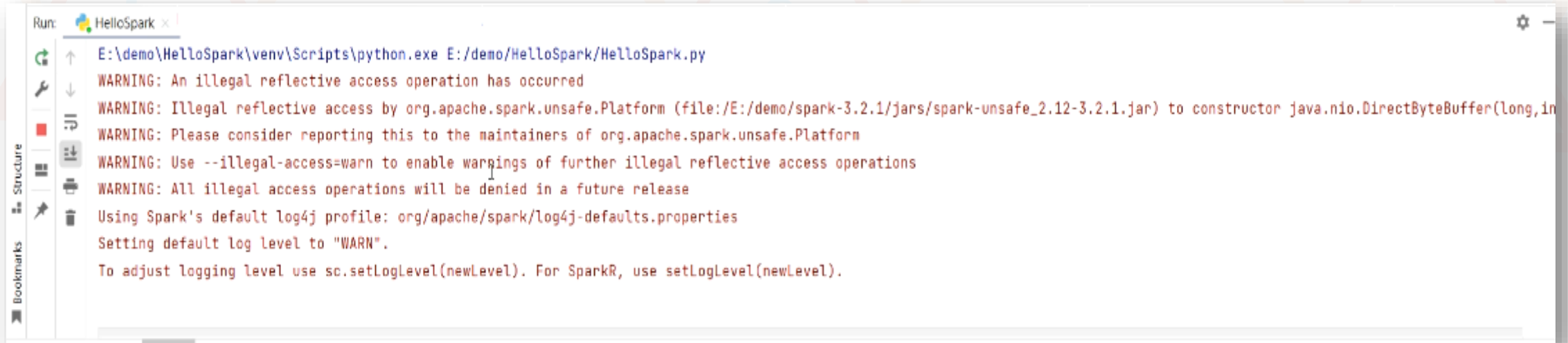
Here is my super simple Spark application. I am creating a Spark Dataframe using a data list. I am giving some column names to list items and executing the show() method on my Dataframe.

Do not stress yourself to understand the code.

A screenshot of a code editor window titled 'HelloSpark.py'. The code is a Python script that initializes a Spark session and creates a DataFrame from a list of tuples. The code is as follows:

```
1 from pyspark.sql import *
2
3 if __name__ == "__main__":
4
5     spark = SparkSession.builder \
6         .appName("Hello Spark") \
7         .master("local[2]") \
8         .getOrCreate()
9
10    data_list = [("Ravi", 28),
11                ("David", 45),
12                ("Abdul", 37)]
13
14    df = spark.createDataFrame(data_list).toDF("Name", "Age")
15    df.show()
16
17
```

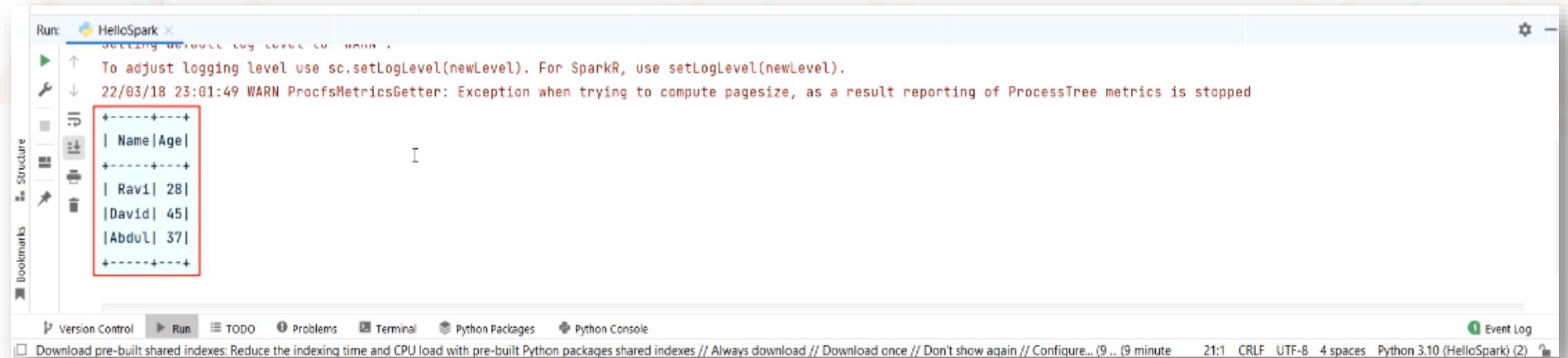
You will see some warnings when you run the code. But you can ignore those warnings.



The screenshot shows a Jupyter Notebook console window titled 'Run: HelloSpark x'. The output displays the command `E:\demo\HelloSpark\venv\Scripts\python.exe E:/demo/HelloSpark/HelloSpark.py` and several warning messages from Spark's unsafe platform. The warnings include: 'An illegal reflective access operation has occurred', 'Illegal reflective access by org.apache.spark.unsafe.Platform (file:/E:/demo/spark-3.2.1/jars/spark-unsafe_2.12-3.2.1.jar) to constructor java.nio.DirectByteBuffer(long,int)', 'Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform', 'Use --illegal-access=warn to enable warnings of further illegal reflective access operations', and 'All illegal access operations will be denied in a future release'. It also shows Spark's default log4j profile and log level settings.

```
Run: HelloSpark x
E:\demo\HelloSpark\venv\Scripts\python.exe E:/demo/HelloSpark/HelloSpark.py
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.spark.unsafe.Platform (file:/E:/demo/spark-3.2.1/jars/spark-unsafe_2.12-3.2.1.jar) to constructor java.nio.DirectByteBuffer(long,int)
WARNING: Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
```

Finally, you will see Dataframe output.
We managed to run a super simple Spark application on our local computer.



The screenshot shows a Jupyter Notebook interface with a 'Run' button and a 'HelloSpark' tab. The output area displays a DataFrame with three rows of data. The DataFrame is highlighted with a red box. The output also includes a warning message about ProcfsMetricsGetter.

```
Setting default log level to 'WARN'.  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
22/03/18 23:01:49 WARN ProcfsMetricsGetter: Exception when trying to compute pagesize, as a result reporting of ProcessTree metrics is stopped
```

Name	Age
Ravi	28
David	45
Abdul	37

The interface also shows a sidebar with 'Structure' and 'Bookmarks' tabs, and a bottom status bar with various settings like 'Version Control', 'Run', 'TODO', 'Problems', 'Terminal', 'Python Packages', 'Python Console', and 'Event Log'.



Thank You
ScholarNest Technologies Pvt Ltd.
www.scholarnest.com